Product Requirements Document (PRD)

CAMPUS CONNECT

Version: v1.0

Status: Draft — Ready for stakeholder review

Document Type: MVP PRD (Discovery & Definition)

I. Executive Summary

Campus Connect is an AI-powered platform that solves the fragmented way students manage their academic and social lives outside of LMS. By consolidating campus information and intelligently forming study groups, it reduces stress, improves collaboration, and fosters community. The primary business goal is to validate the AI-driven Matchmaker feature as the core differentiator for student engagement.

Driven Study Group Matchmaker. The primary business goal is to reduce academic stress and increase student engagement by enabling faster, more relevant peer connections on campus.

Problem: In a Engineering college students face fragmented communication and inefficient organization; forming study groups is manual and chance-driven, causing missed opportunities, stress, and isolation.

Solution: Campus Connect — unified platform with an AI Study Group Matchmaker, integrated AI Academic & Event Calendar, Group Bootstrap with chat deep-links (WhatsApp/Discord), and Discovery for clubs/fests. Manual search & cascading filters are co-located with AI ranking for transparency and control.

Impact: Increase activation and early collaboration: target ≥40% match→join conversion, D28 retention +10pp for matched users, and reduced time-to-first-session.

Investment: 6–8 engineers, 1–2 designers, 1 PM, 1 data engineer; pilot-ready in 12 weeks (MVP), full campus launch in 8–10 months.

II. Target User & Persona

Primary Users • Undergraduate Students (18–24) across STEM/Non-STEM; size per campus: 5k–30k. • Characteristics: mobile-first, use WhatsApp/Discord, need fast academic coordination.

Secondary Users • Final-year students / placement prep cohorts — need placement study groups. • Faculty / Tutors — assignable to groups, mentor support.

Persona: Anjali, the Freshman

- **Age:** 18
- **Background:** First-year Computer Science student, recently moved to campus.
- **Motivations:** Wants to succeed academically, make friends, and feel part of the campus community.
- **Tech Habits:** Heavy WhatsApp/Instagram user, checks LMS for assignments, but finds it clunky. Prefers mobile apps with simple UI.
- **Pain Points:**
    - Struggles to find peers at her pace for study groups.
    - Misses campus events because info is scattered across multiple platforms.
    - Feels isolated when group formation relies on chance.

III. Solution Overview (MVP)

Core MVP Feature: AI-Driven Study Group Matchmaker

- **Inputs:** Student profile (courses, preferred study times, topics needing help, learning style).
- **Process:** AI algorithm matches students with complementary needs and availability.
- **Outputs:** Curated list of potential group members + suggested meeting times.
- **Minimal Feature Set:**
  - Profile creation (basic academic + availability info).
  - AI matchmaker engine.
  - Group creation + notification.
  - Simple chat for coordination.

This MVP validates whether AI-driven matching reduces friction in forming effective study groups.

Clarification: In the "Minimal Feature Set" , clarify that the **Simple Chat** is for *initial* coordination, while **Deep-links** (WhatsApp/Discord) are the secondary "Hand-off" points for long-term study.

IV. User Stories & Acceptance Criteria

User Story 1

As Anjali, I want to create a study profile so that the AI can match me with relevant peers.

- **GIVEN the user has signed up and authenticated via University SSO**
- **WHEN the user enters course and availability details**
- **THEN the system saves the profile and confirms it is editable**

User Story 2

As Anjali, I want to be matched with peers studying the same course so that I can collaborate effectively.

- **GIVEN the user has a completed study profile**
- **WHEN the AI Matchmaker runs**
- **THEN the system suggests at least 3 peers with overlapping courses**

User Story 3

As Anjali, I want to receive notifications when a study group is formed so that I don't miss opportunities.

- **GIVEN the AI has successfully formed a study group**
- **WHEN the group is created**
- **THEN the user receives a push/email notification with group details within 10 seconds**

User Story 4

As Anjali, I want to chat with my matched group so that we can coordinate study sessions.

- **GIVEN the user has accepted a group match**
- **WHEN the group is created**
- **THEN a group chat is auto-generated and messages are delivered in <2 seconds**

User Story 5

As Anjali, I want to update my availability so that the AI can reschedule sessions.

- **GIVEN the user has an active study profile**
- **WHEN the user changes availability preferences**
- **THEN the AI updates matches and suggests new times within 1 minute**

User Story 6

As Anjali, I want to leave a group if it doesn't fit so that I can find better matches.

- **GIVEN the user is part of a study group**
- **WHEN the user selects "Leave Group"**
- **THEN the system removes them and rematches within 24 hours**

User Story 7

As Anjali, I want to see a dashboard of my study groups so that I can track my commitments.

- **GIVEN the user has joined at least one group**
- **WHEN the user opens the dashboard**
- **THEN the system displays active groups, upcoming sessions, and past activity**

User Story 8

As a student, I want to report or block a user so that I can maintain a safe study environment."

- **GIVEN a user is in a group chat.**
- **WHEN they click "Report," the system logs the incident and notifies an admin.**
- **THEN the blocked user can no longer see the reporter's profile or join their future groups.**

V. User Flow Diagram (Step-by-Step)

Flow:

1. App Sign-Up →
2. SSO via University Auth →

3. Create Study Profile (courses, availability, preferences) →
4. AI Matchmaker runs →
5. Suggested group displayed →
6. User accepts →
7. Group chat auto-created →
8. Study session scheduled →
9. Notifications + dashboard tracking.

VI. Technical Requirements

- **AI Inputs:** Course enrollment, preferred study times, topics needing help, learning style.
- **AI Outputs:** Curated list of potential group members + suggested meeting times.
- **Core Components:**
  - AI matching algorithm (collaborative filtering + availability optimization).
  - Notification system (push/email).
  - Group chat module.
- **Third-Party APIs:**
  - University SSO/Auth API.
  - Calendar sync (Google/Outlook)

1. Frontend (User Interface Layer)

  - **Platforms:** Web (React.js), Mobile (React Native or Flutter for cross-platform).
  - **Core Components:**
  - **Onboarding Flow:** Sign-up, University SSO, profile creation.
  - **Dashboard:** Shows study groups, sessions, notifications.
  - **Group Chat UI:** Lightweight messaging interface with basic features (text, file sharing).

- **Notifications:** Push + in-app alerts for group formation and session scheduling.
- **Accessibility:** WCAG 2.1 compliance, multilingual support (English, Hindi, Urdu).
- **Performance:** <2s load time for dashboard and group match results.

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2. Backend (Application Logic Layer)

- **Frameworks:** Node.js / Express or Django (Python).
- **Core Services:**
- **Authentication Service:** University SSO integration (OAuth2, SAML).
- **User Profile Service:** Stores academic info, availability, preferences.
- **Matchmaker Service:** Calls AI/ML engine to generate group recommendations.
- **Notification Service:** Push/email triggers via Firebase/APNs.
- **Chat Service:** Real-time messaging (WebSockets or Firebase Realtime DB).
- **APIs:** REST/GraphQL endpoints for frontend consumption.
- **Scalability:** Microservices architecture to support 10k+ concurrent users.

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3. AI/ML Component (Matchmaker Engine)

- **Inputs:** Course enrollment, preferred study times, topics needing help, learning style.
- **Outputs:** Curated list of potential group members + suggested meeting times.
- **Algorithm Approaches:**
- Collaborative filtering (peer similarity).
- Constraint-based optimization (availability overlap).
- Sentiment/feedback loop (to refine match quality).

- **Data Storage:** Training data anonymized, stored securely.
- **Performance:** Match results generated in <2s.

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4. Integrations

- **University Systems:** LMS (Canvas, Moodle, Blackboard) for course sync.
- **Calendar APIs:** Google Calendar, Outlook for scheduling.
- **Library Booking API:** Room reservations for study sessions.
- **Notification APIs:** Firebase Cloud Messaging (Android), Apple Push Notification Service (iOS).

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5. Infrastructure & DevOps

- **Hosting:** Cloud-based (AWS, Azure, or GCP).
- **Database:** PostgreSQL/MySQL for structured data; MongoDB for chat logs.
- **Caching:** Redis for fast match queries.
- **Deployment:** Docker + Kubernetes for container orchestration.
- **Monitoring:** Prometheus + Grafana dashboards.
- **Security:** End-to-end encryption for chat, TLS for all API calls, GDPR/FERPA compliance.
- **Reliability:** 99.9% uptime SLA.

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6. Non-Functional Requirements

- **Latency:** <2s for match queries, <1s for chat messages.
- **Reliability:** Auto-scaling to handle peak exam season traffic.
- **Data Privacy:** Opt-in consent for data sharing, anonymized analytics.
- **Localization:** Language packs for regional campus

- **Logic Addition:** Add a sub-bullet under "Algorithm Approaches" called **"Fallback Logic."**

- • **Requirement:** If the algorithm returns zero matches, the system must trigger a "Waitlist" state instead of an empty screen. This ensures the user doesn't feel the app is "broken."
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VII. Launch Success Metrics (KPIs)

Mapped to **AARRR Funnel**:

- **Acquisition:** 40% of campus students sign up within first semester.
- **Activation:** 70% of users create a study profile within 1 week of sign-up.
- **Retention:** 50% of active users participate in at least 2 study groups per month.
- **Referral (optional stretch):** 20% of users invite peers to join via referral links.

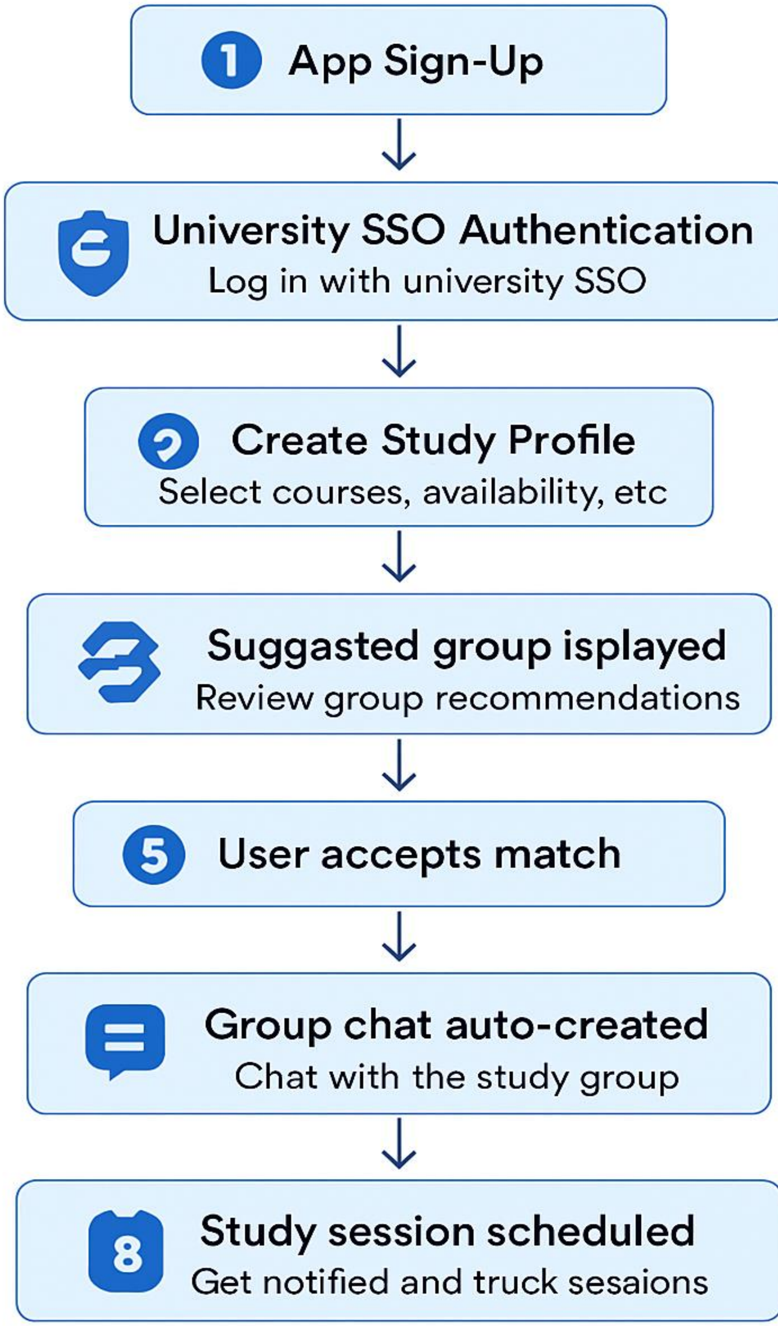
Funnel Stage	Metric	Definition	Goal
Acquisition	Sign-Up Rate	% of total campus students who register on the app	≥ 40% in first semester
Activation	Profile Completion Rate	% of users who create a full study profile (courses, availability, preferences)	≥ 70% within 7 days
Retention	Group Participation Rate	% of users who join at least 2 study groups per month	≥ 50% monthly
Retention	Session Attendance Rate	% of scheduled study sessions that are attended by matched users	≥ 80%
Referral	Peer Invite Rate	% of users who invite others via referral links	≥ 20%
Engagement (Bonus)	Match Satisfaction Score	Avg. rating of AI match quality (1–5 scale)	≥ 4.2/5

Why These KPIs Matter

- Sign-Up Rate shows initial interest and marketing effectiveness.

- Profile Completion validates onboarding UX and user motivation.
- Group Participation & Attendance prove the AI Matchmaker is solving the core problem.
- Referral Rate indicates organic growth and peer trust.
- Satisfaction Score helps refine the matching algorithm and UX.

User flow Diagram



Study Match

AI-Driven Study coordination